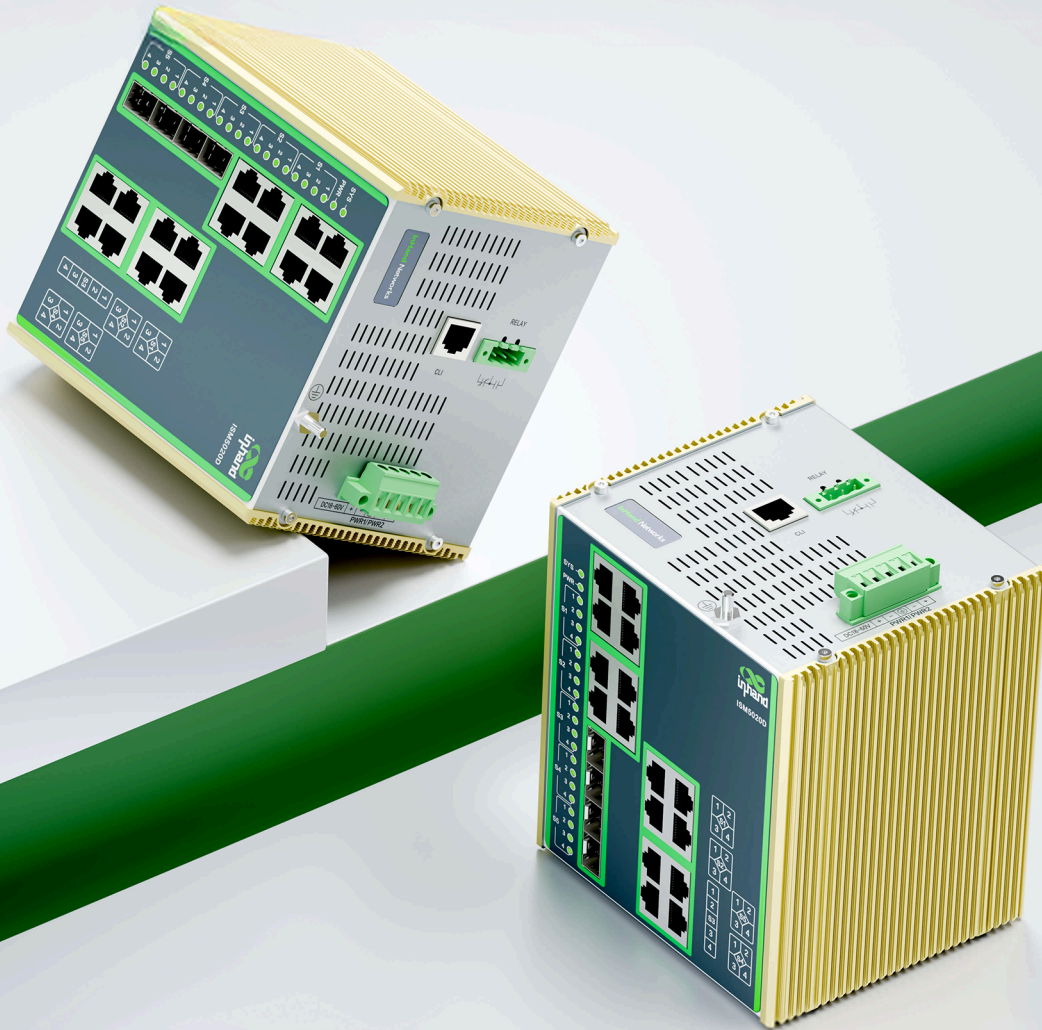


Build advanced and highly reliable industrial Ethernet communication system



ISM5020D Managed Industrial Ethernet Switch

· Public Utility

· Smart Manufacturing

· Smart City

· Smart Energy

1. Product Overview

The ISM series Managed Industrial Ethernet Switch has been purpose-built to cater to the rigorous demands of industrial settings, including power, transportation, and industrial control applications.

Product Features:

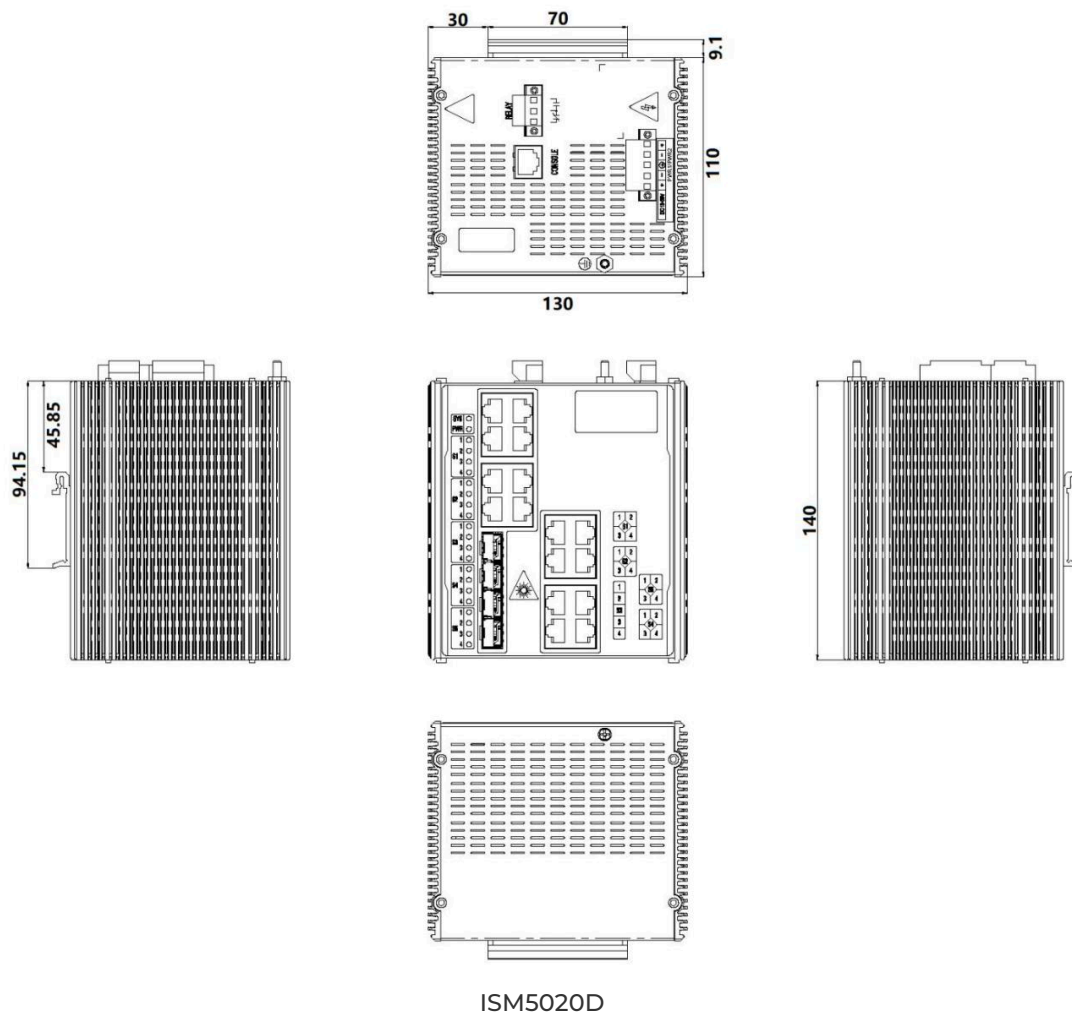
- **Reliability & Resilience:** Fanless design, IP40 protection rating, rugged metal casing, dust and dirt resistance, and wide temperature operation (-40 °C ~ +85 °C (-40 °F ~ +185 °F)). Industrial-grade redundant wide-voltage power supply.
- **Redundancy Protocols:** Supports RSTP / MSTP / ERPS / MRP ring redundancy protocols (recovery time ≤50ms).
- **Network Management:** Supports SNMP v2c/v3 for integrated network management and RMON for effective network monitoring. Allows graphical representation of traffic, device temperature, CPU usage, and neighbor devices.
- **Enhanced Security:** Dynamic ARP Inspection, DHCP Snooping, L2/L3/L4 ACLs, IP Source Guard, overvoltage protection, IGMP Snooping, and VLAN segmentation.
- **Advanced Control:** Dying gasp power-down reporting, WEB/CLI factory reset scripts, 8 ToS/DSCP QoS queues, sFlow protocol, and static routing.

Key Technical Specifications

Parameter	Specification
Type	Layer 3 Managed Industrial Ethernet Switch
Ports	16 x 10/100/1000BaseT + 4 x 100/1000/2500BaseX SFP
Switching Performance	68 Gbps backplane bandwidth; 16K MAC table; 1G RAM
Dimensions / Weight	130 mm x 140 mm x 110 mm; 1.3 kg (2.87 lb)

Parameter	Specification
Power	18~60 VDC redundant dual input
Environment	-40 °C ~ +85 °C (-40 °F ~ +185 °F) operating; IP40
EMC	FCC Part 15 Class A; IEC 61000-4-2/3/4/5/6/8/11/12
Management	Web, CLI, SNMPv1/v2c/v3, RMON; ERPS/MRP/RSTP/MSTP, IGMP, VLAN, QoS 8 queues
Certifications	CE, FCC, IEC61850-3

2. Product Dimensions



Note:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.

2. All dimensions are approximate and **for reference only**.
3. The dimensions shown in the figure **shall not be used for production or processing**.
4. Dimensions must comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

3. Technical Specifications

3.1 Protocol Compliance List

Category/Parameter	Specification
IEEE Standards	
IEEE 802.3	CSMA/CD method and physical Layer specifications
IEEE 802.1p	Class of Service
IEEE 802.1q	VLAN tagging
IEEE 802.1d	Spanning Tree Algorithm
IEEE 802.1w	Rapid Spanning Tree
IEEE 802.1s	Multiple Spanning Tree
IEEE 802.3ac	VLAN Tagging
IEEE 802.1x	Authentication
IEEE 802.3ad	Link Aggregation
IEEE 802.3x	Flow Control
IEEE 802.3	Ethernet
IEEE 802.3u	Fast Ethernet
IEEE 802.3z	Gigabit Ethernet

Category/Parameter	Specification
IEEE 802.1ab	Link Layer Discovery Protocol
RFC Standards	
RFC 768	UDP
RFC 791	IP
RFC 792	ICMP
RFC 793	TCP
RFC 826	ARP
RFC 854	Telnet Client & Server
RFC 862	Echo Protocol
RFC 863	Discard Protocol
RFC 904	Exterior Gateway Protocol Formal Specification
RFC 1027	Using ARP to Implement Transparent Subnet Gateways
RFC 1058	RIP
RFC 1059, 1119	NTPv1/2
RFC 1112	IGMP
RFC 1191	Path MTU Discovery
RFC 1256	ICMP Router discovery protocol
RFC 1267	A Border Gateway Protocol 3 (BGP-3)
RFC 1388	RIP Version 2 Carrying Additional Information
RFC 1403	BGP OSPF Interaction
RFC 1519	CIDR (Classless Inter-domain Routing)

Category/Parameter	Specification
RFC 1587	OSPF NSSA
RFC 1812	Requirements for IP Version 4 Routers
RFC 1994	PPP Challenge Handshake Authentication Protocol (CHAP)
RFC 2068	HTTP
RFC 213	DHCP Server
RFC 2138	RADIUS
RFC 2139	RADIUS Accounting
RFC 2236	IGMPv2
RFC 2328	OSPF V2
RFC 2338	VRRP
RFC 2370	The OSPF Opaque LSA Option
RFC 2474	DiffServ Precedence
RFC 2475	DiffServ Core and Edge Router Functions
RFC 2597	DiffServ Assured Forwarding
RFC 2598	DiffServ Expedited Forwarding
RFC 2644	Directed Broadcasts
RFC 2865	Remote Authentication Dial User Service (RADIUS)
RFC 3046	DHCP Relay Agent Information Option
RFC 3222	Forwarding Information Base (FIB)
Other Protocols	
GMRP	GARP

Category/Parameter	Specification
GVRP	GARP
SSH2	Secure Shell 2
IGMP	Snooping
SNMPv3	Supported

3.2 Hardware Specifications

Category/Parameter	Specification
Physical Performance	
Enclosure	Fully enclosed seamless metal enclosure
Dimensions (W × D × H)	130 mm × 140 mm × 110 mm
Weight	1.3 kg (2.87 lb)
Mounting Method	DIN-rail mounting
Cooling Method	Fanless cooling
Ingress Protection	IP40
Storage Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Operating Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Humidity	5 ~ 95% (non-condensing)
Hardware Performance	
Backplane Bandwidth	68 Gbps
Transmission Mode	Parallel Storage Forwarding

Category/Parameter	Specification
RAM	1G
Flash	32M
MAC Table Size	16K
Packet Buffer Size	4 Mbits
Exchange Rate	148,800 pps/100M ports; 1,488,000 pps/1000M ports
Power Parameters	
Input Voltage	18-60 VDC Redundant dual input
Overload Current Protection	Supported
Reverse Polarity Protection	Supported
Electromagnetic Characteristics	
EMI	FCC 47 CFR Part 15 Class A; EN55022 Class A
EMS	IEC(EN)61000-4-2, Class 4 IEC(EN)61000-4-3, Class 3 IEC(EN)61000-4-4, Class 3 IEC(EN)61000-4-5, Class 4 IEC(EN)61000-4-6, Class 3 IEC(EN)61000-4-8, Class 5 IEC(EN)61000-4-11, Class 4 IEC(EN)61000-4-12, Class 3
Mechanical Characteristics	
Shock	IEC60068-2-27
Freefall	IEC60068-2-31

Category/Parameter	Specification
Vibration	IEC60068-2-6
Certifications	
Certifications	CE, FCC, IEC61850-3
Quality Assurance	
Warranty Period	5 years
MTBF	35 years

4. Software Specifications

Software Functions	Specification
Redundancy	ERPS, MRP, RSTP, MSTP, Port Trunking
Management Mode	Web, serial port, STD-17 MIB-II, STD-58 SMIV2, STD-59 RMON, STD-62 SNMPv3, SNMPv2c, SNMPv1, RFC2925 Ping MIB
Time Synchronization	SNTP
Diagnostic Mode	Indicator light, journal file, relay, RMON, port mirroring, TRAP
Others	4K VLANs ID, 256 Active VLANs, IPv4/IPv6 multicast, storm control, MC/BC protection, support Jumbo Frame, HTTPS, SSHv2, and SFTP, supports QoS 8 queues

5. Ordering Guide

Model	Description
ISM5020D-P-4GSFP-16GT-24	20-port Layer 3 managed Industrial Switch. 16 <i>10/100/1000Base-T/TX RJ45 Autosense MDI/MDI-X Port + 4 100/1000/2500BaseX SFP slots</i> . Isolated Dual 18-60VDC Power Inputs. 1 Management Serial CLI port, 1 Alarm Relay port.

6. Contact Us

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